

Profit Maximization Practice Set

Instructions: For each exercise below, you are given a firm's Total Revenue and Total Cost functions in terms of output Q . Use differentiation to find the profit-maximizing output level, then calculate Total Revenue, Total Cost, and Profit at that level of output. Each exercise carries 10 marks.

Exercise 1

Given:

Total Revenue: $TR(Q) = 80Q - 2Q^2$

Total Cost: $TC(Q) = 20Q + 100$

Tasks:

1. Write down the profit function: $\pi(Q) = TR(Q) - TC(Q)$.
2. Find $d\pi/dQ$, set it equal to zero, and solve for Q^* (the profit-maximizing output level).
3. Calculate Total Revenue (TR) at $Q = Q^*$.
4. Calculate Total Cost (TC) at $Q = Q^*$.
5. Calculate Profit (π) at $Q = Q^*$, and verify it equals $TR - TC$ from your previous steps.

Exercise 2

Given:

Total Revenue: $TR(Q) = 100Q - 2Q^2$

Total Cost: $TC(Q) = Q^2 + 40Q + 80$

Tasks:

1. Write down the profit function: $\pi(Q) = TR(Q) - TC(Q)$.
2. Find $d\pi/dQ$, set it equal to zero, and solve for Q^* (the profit-maximizing output level).
3. Calculate Total Revenue (TR) at $Q = Q^*$.
4. Calculate Total Cost (TC) at $Q = Q^*$.
5. Calculate Profit (π) at $Q = Q^*$, and verify it equals $TR - TC$ from your previous steps.

Exercise 3

Given:

Total Revenue: $TR(Q) = 150Q - 3Q^2$

Total Cost: $TC(Q) = 30Q + 200$

Tasks:

1. Write down the profit function: $\pi(Q) = TR(Q) - TC(Q)$.

2. Find $d\pi/dQ$, set it equal to zero, and solve for Q^* (the profit-maximizing output level).
3. Calculate Total Revenue (TR) at $Q = Q^*$.
4. Calculate Total Cost (TC) at $Q = Q^*$.
5. Calculate Profit (π) at $Q = Q^*$, and verify it equals $TR - TC$ from your previous steps.

Exercise 4

Given:**Total Revenue:** $TR(Q) = 180Q - 2Q^2$ **Total Cost:** $TC(Q) = 3Q^2 + 60Q + 90$ **Tasks:**

1. Write down the profit function: $\pi(Q) = TR(Q) - TC(Q)$.
2. Find $d\pi/dQ$, set it equal to zero, and solve for Q^* (the profit-maximizing output level).
3. Calculate Total Revenue (TR) at $Q = Q^*$.
4. Calculate Total Cost (TC) at $Q = Q^*$.
5. Calculate Profit (π) at $Q = Q^*$, and verify it equals $TR - TC$ from your previous steps.

Exercise 5

Given:**Total Revenue:** $TR(Q) = 250Q - 3Q^2$ **Total Cost:** $TC(Q) = Q^2 + 50Q + 400$ **Tasks:**

1. Write down the profit function: $\pi(Q) = TR(Q) - TC(Q)$.
2. Find $d\pi/dQ$, set it equal to zero, and solve for Q^* (the profit-maximizing output level).
3. Calculate Total Revenue (TR) at $Q = Q^*$.
4. Calculate Total Cost (TC) at $Q = Q^*$.
5. Calculate Profit (π) at $Q = Q^*$, and verify it equals $TR - TC$ from your previous steps.

Answer Key

Each solution below is worked in full, step by step, with the reasoning behind every calculation — the level of detail expected for a 10-mark question.

Exercise 1

$$TR(Q) = 80Q - 2Q^2 \quad TC(Q) = 20Q + 100$$

Step 1: Derive the Profit Function

Profit is defined as the excess of total revenue over total cost. We obtain $\pi(Q)$ by subtracting the cost function from the revenue function and collecting like terms.

$$\begin{aligned}\pi(Q) &= TR(Q) - TC(Q) \\ &= (80Q - 2Q^2) - (20Q + 100) \\ &= 80Q - 2Q^2 - 20Q - 100 \\ \pi(Q) &= -2Q^2 + 60Q - 100\end{aligned}$$

Step 2: Apply the First-Order Condition (FOC)

A function reaches a maximum (or minimum) where its slope is zero. We therefore differentiate $\pi(Q)$ with respect to Q and set the result equal to zero — this is equivalent to the standard rule that profit is maximized where Marginal Revenue equals Marginal Cost ($MR = MC$).

$$\begin{aligned}d\pi/dQ &= -4Q + 60 \\ \text{Setting } d\pi/dQ &= 0: \\ -4Q + 60 &= 0 \\ 4Q &= 60 \\ Q^* &= 15\end{aligned}$$

Step 3: Verify the Second-Order Condition (SOC)

The FOC alone only identifies a turning point; it could be a maximum or a minimum. To confirm it is a maximum, we take the second derivative of $\pi(Q)$. A negative value indicates the profit curve is concave (bends downward), which is the condition for a maximum.

$$d^2\pi/dQ^2 = -4$$

Since $d^2\pi/dQ^2 = -4$ is negative, the profit function is concave (an inverted U-shape) at this point. This confirms that $Q^* = 15$ is a point of maximum profit, not minimum profit.

Step 4: Calculate Total Revenue at Q^*

Substitute Q^* back into the original Total Revenue function to find the revenue earned at the profit-maximizing output level.

$$\begin{aligned}TR(15) &= 80(15) - 2(15)^2 \\ &= 1200 - 2(225) \\ &= 1200 - 450 \\ TR &= 750\end{aligned}$$

Step 5: Calculate Total Cost at Q^*

Substitute Q^* back into the original Total Cost function to find the cost incurred at the profit-maximizing output level.

$$\begin{aligned}
 TC(15) &= 20(15) + 100 \\
 &= 300 + 100 \\
 \mathbf{TC} &= \mathbf{400}
 \end{aligned}$$

Step 6: Calculate Profit at Q^* and Cross-Check

Profit at Q^* can be found in two independent ways: (a) by subtracting TC from TR, and (b) by substituting Q^* directly into the profit function $\pi(Q)$ derived in Step 1. Both methods must give the same answer; this cross-check guards against arithmetic errors.

$$\begin{aligned}
 \pi(15) &= TR(15) - TC(15) \\
 &= 750 - 400 \\
 \mathbf{\pi} &= \mathbf{350}
 \end{aligned}$$

Cross-check using $\pi(Q)$:

$$\begin{aligned}
 \pi(15) &= -2(15)^2 + 60(15) - 100 \\
 &= -450 + 900 - 100 \\
 &= 350 \quad \checkmark \text{ matches Step 6}
 \end{aligned}$$

The firm maximizes profit at an output level of $Q^* = 15$ units. At this level it earns Total Revenue of 750, incurs Total Cost of 400, and realizes a maximum Profit of 350. Both methods of calculating profit ($TR - TC$, and direct substitution into $\pi(Q)$) agree, confirming the answer is internally consistent.

Q^* 15	Total Revenue 750	Total Cost 400	Maximum Profit 350
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Marking Scheme (10 marks)

Marking Criteria	Marks
Correctly derive and simplify the profit function $\pi(Q) = TR(Q) - TC(Q)$	2
Differentiate $\pi(Q)$ correctly, set $d\pi/dQ = 0$, and solve correctly for Q^*	3
State the second-order condition and correctly confirm it is a maximum	1
Correctly calculate Total Revenue at $Q = Q^*$	1
Correctly calculate Total Cost at $Q = Q^*$	1
Correctly calculate Profit at $Q = Q^*$, with cross-check and valid conclusion	2
Total	10

Exercise 2

$$TR(Q) = 100Q - 2Q^2 \quad TC(Q) = Q^2 + 40Q + 80$$

Step 1: Derive the Profit Function

Profit is defined as the excess of total revenue over total cost. We obtain $\pi(Q)$ by subtracting the cost function from the revenue function and collecting like terms.

$$\begin{aligned}\pi(Q) &= TR(Q) - TC(Q) \\ &= (100Q - 2Q^2) - (Q^2 + 40Q + 80) \\ &= 100Q - 2Q^2 - Q^2 - 40Q - 80 \\ \pi(Q) &= -3Q^2 + 60Q - 80\end{aligned}$$

Step 2: Apply the First-Order Condition (FOC)

A function reaches a maximum (or minimum) where its slope is zero. We therefore differentiate $\pi(Q)$ with respect to Q and set the result equal to zero — this is equivalent to the standard rule that profit is maximized where Marginal Revenue equals Marginal Cost ($MR = MC$).

$$\begin{aligned}d\pi/dQ &= -6Q + 60 \\ \text{Setting } d\pi/dQ &= 0: \\ -6Q + 60 &= 0 \\ 6Q &= 60 \\ Q^* &= 10\end{aligned}$$

Step 3: Verify the Second-Order Condition (SOC)

The FOC alone only identifies a turning point; it could be a maximum or a minimum. To confirm it is a maximum, we take the second derivative of $\pi(Q)$. A negative value indicates the profit curve is concave (bends downward), which is the condition for a maximum.

$$d^2\pi/dQ^2 = -6$$

Since $d^2\pi/dQ^2 = -6$ is negative, the profit function is concave at this point, confirming $Q^* = 10$ is a maximum.

Step 4: Calculate Total Revenue at Q^*

Substitute Q^* back into the original Total Revenue function to find the revenue earned at the profit-maximizing output level.

$$\begin{aligned}TR(10) &= 100(10) - 2(10)^2 \\ &= 1000 - 2(100) \\ &= 1000 - 200 \\ TR &= 800\end{aligned}$$

Step 5: Calculate Total Cost at Q^*

Substitute Q^* back into the original Total Cost function to find the cost incurred at the profit-maximizing output level.

$$\begin{aligned}TC(10) &= (10)^2 + 40(10) + 80 \\ &= 100 + 400 + 80 \\ TC &= 580\end{aligned}$$

Step 6: Calculate Profit at Q^* and Cross-Check

Profit at Q^* can be found in two independent ways: (a) by subtracting TC from TR, and (b) by substituting Q^* directly into the profit function $\pi(Q)$ derived in Step 1. Both methods must give the same answer; this cross-check guards against arithmetic errors.

$$\begin{aligned}\pi(10) &= TR(10) - TC(10) \\ &= 800 - 580 \\ \pi &= 220\end{aligned}$$

Cross-check using $\pi(Q)$:

$$\begin{aligned}\pi(10) &= -3(10)^2 + 60(10) - 80 \\ &= -300 + 600 - 80 \\ &= 220 \quad \checkmark \text{ matches Step 6}\end{aligned}$$

Conclusion:

The firm maximizes profit at an output level of $Q^* = 10$ units, earning Total Revenue of 800 against Total Cost of 580, for a maximum Profit of 220. The agreement between the two calculation routes confirms the result is correct.

Q^* 10	Total Revenue 800	Total Cost 580	Maximum Profit 220
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Marking Scheme (10 marks)

Marking Criteria	Marks
Correctly derive and simplify the profit function $\pi(Q) = TR(Q) - TC(Q)$	2
Differentiate $\pi(Q)$ correctly, set $d\pi/dQ = 0$, and solve correctly for Q^*	3
State the second-order condition and correctly confirm it is a maximum	1
Correctly calculate Total Revenue at $Q = Q^*$	1
Correctly calculate Total Cost at $Q = Q^*$	1
Correctly calculate Profit at $Q = Q^*$, with cross-check and valid conclusion	2
Total	10

Exercise 3

$$TR(Q) = 150Q - 3Q^2 \quad TC(Q) = 30Q + 200$$

Step 1: Derive the Profit Function

Profit is defined as the excess of total revenue over total cost. We obtain $\pi(Q)$ by subtracting the cost function from the revenue function and collecting like terms.

$$\begin{aligned}\pi(Q) &= TR(Q) - TC(Q) \\ &= (150Q - 3Q^2) - (30Q + 200) \\ &= 150Q - 3Q^2 - 30Q - 200 \\ \pi(Q) &= -3Q^2 + 120Q - 200\end{aligned}$$

Step 2: Apply the First-Order Condition (FOC)

A function reaches a maximum (or minimum) where its slope is zero. We therefore differentiate $\pi(Q)$ with respect to Q and set the result equal to zero — this is equivalent to the standard rule that profit is maximized where Marginal Revenue equals Marginal Cost ($MR = MC$).

$$\begin{aligned}d\pi/dQ &= -6Q + 120 \\ \text{Setting } d\pi/dQ &= 0: \\ -6Q + 120 &= 0 \\ 6Q &= 120 \\ Q^* &= 20\end{aligned}$$

Step 3: Verify the Second-Order Condition (SOC)

The FOC alone only identifies a turning point; it could be a maximum or a minimum. To confirm it is a maximum, we take the second derivative of $\pi(Q)$. A negative value indicates the profit curve is concave (bends downward), which is the condition for a maximum.

$$d^2\pi/dQ^2 = -6$$

Since $d^2\pi/dQ^2 = -6$ is negative, the profit function is concave at this point, confirming $Q^* = 20$ is a maximum.

Step 4: Calculate Total Revenue at Q^*

Substitute Q^* back into the original Total Revenue function to find the revenue earned at the profit-maximizing output level.

$$\begin{aligned}TR(20) &= 150(20) - 3(20)^2 \\ &= 3000 - 3(400) \\ &= 3000 - 1200 \\ TR &= 1800\end{aligned}$$

Step 5: Calculate Total Cost at Q^*

Substitute Q^* back into the original Total Cost function to find the cost incurred at the profit-maximizing output level.

$$\begin{aligned}TC(20) &= 30(20) + 200 \\ &= 600 + 200 \\ TC &= 800\end{aligned}$$

Step 6: Calculate Profit at Q^* and Cross-Check

Profit at Q^* can be found in two independent ways: (a) by subtracting TC from TR, and (b) by substituting Q^* directly into the profit function $\pi(Q)$ derived in Step 1. Both methods must give the same answer; this cross-check guards against arithmetic errors.

$$\begin{aligned}\pi(20) &= TR(20) - TC(20) \\ &= 1800 - 800 \\ \pi &= 1000\end{aligned}$$

Cross-check using $\pi(Q)$:

$$\begin{aligned}\pi(20) &= -3(20)^2 + 120(20) - 200 \\ &= -1200 + 2400 - 200 \\ &= 1000 \quad \checkmark \text{ matches Step 6}\end{aligned}$$

Conclusion:

The firm maximizes profit at an output level of $Q^* = 20$ units, earning Total Revenue of 1800 against Total Cost of 800, for a maximum Profit of 1000. The two calculation routes agree, confirming the result.

Q^* 20	Total Revenue 1800	Total Cost 800	Maximum Profit 1000
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Marking Scheme (10 marks)

Marking Criteria	Marks
Correctly derive and simplify the profit function $\pi(Q) = TR(Q) - TC(Q)$	2
Differentiate $\pi(Q)$ correctly, set $d\pi/dQ = 0$, and solve correctly for Q^*	3
State the second-order condition and correctly confirm it is a maximum	1
Correctly calculate Total Revenue at $Q = Q^*$	1
Correctly calculate Total Cost at $Q = Q^*$	1
Correctly calculate Profit at $Q = Q^*$, with cross-check and valid conclusion	2
Total	10

Exercise 4

$$TR(Q) = 180Q - 2Q^2 \quad TC(Q) = 3Q^2 + 60Q + 90$$

Step 1: Derive the Profit Function

Profit is defined as the excess of total revenue over total cost. We obtain $\pi(Q)$ by subtracting the cost function from the revenue function and collecting like terms.

$$\begin{aligned}\pi(Q) &= TR(Q) - TC(Q) \\ &= (180Q - 2Q^2) - (3Q^2 + 60Q + 90) \\ &= 180Q - 2Q^2 - 3Q^2 - 60Q - 90 \\ \pi(Q) &= -5Q^2 + 120Q - 90\end{aligned}$$

Step 2: Apply the First-Order Condition (FOC)

A function reaches a maximum (or minimum) where its slope is zero. We therefore differentiate $\pi(Q)$ with respect to Q and set the result equal to zero — this is equivalent to the standard rule that profit is maximized where Marginal Revenue equals Marginal Cost ($MR = MC$).

$$\begin{aligned}d\pi/dQ &= -10Q + 120 \\ \text{Setting } d\pi/dQ &= 0: \\ -10Q + 120 &= 0 \\ 10Q &= 120 \\ Q^* &= 12\end{aligned}$$

Step 3: Verify the Second-Order Condition (SOC)

The FOC alone only identifies a turning point; it could be a maximum or a minimum. To confirm it is a maximum, we take the second derivative of $\pi(Q)$. A negative value indicates the profit curve is concave (bends downward), which is the condition for a maximum.

$$d^2\pi/dQ^2 = -10$$

Since $d^2\pi/dQ^2 = -10$ is negative, the profit function is concave at this point, confirming $Q^* = 12$ is a maximum.

Step 4: Calculate Total Revenue at Q^*

Substitute Q^* back into the original Total Revenue function to find the revenue earned at the profit-maximizing output level.

$$\begin{aligned}TR(12) &= 180(12) - 2(12)^2 \\ &= 2160 - 2(144) \\ &= 2160 - 288 \\ TR &= 1872\end{aligned}$$

Step 5: Calculate Total Cost at Q^*

Substitute Q^* back into the original Total Cost function to find the cost incurred at the profit-maximizing output level.

$$\begin{aligned}TC(12) &= 3(12)^2 + 60(12) + 90 \\ &= 432 + 720 + 90 \\ TC &= 1242\end{aligned}$$

Step 6: Calculate Profit at Q^* and Cross-Check

Profit at Q^* can be found in two independent ways: (a) by subtracting TC from TR, and (b) by substituting Q^* directly into the profit function $\pi(Q)$ derived in Step 1. Both methods must give the same answer; this cross-check guards against arithmetic errors.

$$\begin{aligned}\pi(12) &= TR(12) - TC(12) \\ &= 1872 - 1242 \\ \pi &= 630\end{aligned}$$

Cross-check using $\pi(Q)$:

$$\begin{aligned}\pi(12) &= -5(12)^2 + 120(12) - 90 \\ &= -720 + 1440 - 90 \\ &= 630 \quad \checkmark \text{ matches Step 6}\end{aligned}$$

Conclusion:

The firm maximizes profit at an output level of $Q^* = 12$ units, earning Total Revenue of 1872 against Total Cost of 1242, for a maximum Profit of 630. The two calculation routes agree, confirming the result.

Q^* 12	Total Revenue 1872	Total Cost 1242	Maximum Profit 630
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Marking Scheme (10 marks)

Marking Criteria	Marks
Correctly derive and simplify the profit function $\pi(Q) = TR(Q) - TC(Q)$	2
Differentiate $\pi(Q)$ correctly, set $d\pi/dQ = 0$, and solve correctly for Q^*	3
State the second-order condition and correctly confirm it is a maximum	1
Correctly calculate Total Revenue at $Q = Q^*$	1
Correctly calculate Total Cost at $Q = Q^*$	1
Correctly calculate Profit at $Q = Q^*$, with cross-check and valid conclusion	2
Total	10

Exercise 5

$$TR(Q) = 250Q - 3Q^2 \quad TC(Q) = Q^2 + 50Q + 400$$

Step 1: Derive the Profit Function

Profit is defined as the excess of total revenue over total cost. We obtain $\pi(Q)$ by subtracting the cost function from the revenue function and collecting like terms.

$$\begin{aligned}\pi(Q) &= TR(Q) - TC(Q) \\ &= (250Q - 3Q^2) - (Q^2 + 50Q + 400) \\ &= 250Q - 3Q^2 - Q^2 - 50Q - 400 \\ \pi(Q) &= -4Q^2 + 200Q - 400\end{aligned}$$

Step 2: Apply the First-Order Condition (FOC)

A function reaches a maximum (or minimum) where its slope is zero. We therefore differentiate $\pi(Q)$ with respect to Q and set the result equal to zero — this is equivalent to the standard rule that profit is maximized where Marginal Revenue equals Marginal Cost ($MR = MC$).

$$\begin{aligned}d\pi/dQ &= -8Q + 200 \\ \text{Setting } d\pi/dQ &= 0: \\ -8Q + 200 &= 0 \\ 8Q &= 200 \\ Q^* &= 25\end{aligned}$$

Step 3: Verify the Second-Order Condition (SOC)

The FOC alone only identifies a turning point; it could be a maximum or a minimum. To confirm it is a maximum, we take the second derivative of $\pi(Q)$. A negative value indicates the profit curve is concave (bends downward), which is the condition for a maximum.

$$d^2\pi/dQ^2 = -8$$

Since $d^2\pi/dQ^2 = -8$ is negative, the profit function is concave at this point, confirming $Q^* = 25$ is a maximum.

Step 4: Calculate Total Revenue at Q^*

Substitute Q^* back into the original Total Revenue function to find the revenue earned at the profit-maximizing output level.

$$\begin{aligned}TR(25) &= 250(25) - 3(25)^2 \\ &= 6250 - 3(625) \\ &= 6250 - 1875 \\ TR &= 4375\end{aligned}$$

Step 5: Calculate Total Cost at Q^*

Substitute Q^* back into the original Total Cost function to find the cost incurred at the profit-maximizing output level.

$$\begin{aligned}TC(25) &= (25)^2 + 50(25) + 400 \\ &= 625 + 1250 + 400 \\ TC &= 2275\end{aligned}$$

Step 6: Calculate Profit at Q^* and Cross-Check

Profit at Q^* can be found in two independent ways: (a) by subtracting TC from TR, and (b) by substituting Q^* directly into the profit function $\pi(Q)$ derived in Step 1. Both methods must give the same answer; this cross-check guards against arithmetic errors.

$$\begin{aligned}\pi(25) &= TR(25) - TC(25) \\ &= 4375 - 2275 \\ \pi &= 2100\end{aligned}$$

Cross-check using $\pi(Q)$:

$$\begin{aligned}\pi(25) &= -4(25)^2 + 200(25) - 400 \\ &= -2500 + 5000 - 400 \\ &= 2100 \quad \checkmark \text{ matches Step 6}\end{aligned}$$

Conclusion:

The firm maximizes profit at an output level of $Q^* = 25$ units, earning Total Revenue of 4375 against Total Cost of 2275, for a maximum Profit of 2100. The two calculation routes agree, confirming the result.

Q^* 25	Total Revenue 4375	Total Cost 2275	Maximum Profit 2100
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Marking Scheme (10 marks)

Marking Criteria	Marks
Correctly derive and simplify the profit function $\pi(Q) = TR(Q) - TC(Q)$	2
Differentiate $\pi(Q)$ correctly, set $d\pi/dQ = 0$, and solve correctly for Q^*	3
State the second-order condition and correctly confirm it is a maximum	1
Correctly calculate Total Revenue at $Q = Q^*$	1
Correctly calculate Total Cost at $Q = Q^*$	1
Correctly calculate Profit at $Q = Q^*$, with cross-check and valid conclusion	2
Total	10